

Auntu Gain

EMBEDDED & AUTONOMOUS SYSTEMS ENGINEER

Research Direction

Engineer with 5+ years of experience developing embedded, autonomous, and industrial systems for real physical environments, including autonomous surface vehicles, vision-based robotic automation, production automotive software, and adaptive edge anomaly detection.

Graduate research interest: reliable decision-making in deployed robotic and industrial systems, particularly how perception and condition-monitoring models should respond to changing operating conditions and abnormal data, when a system should act or abstain, and how adaptive models can avoid absorbing abnormal behavior into their definition of normal. I want to study these questions systematically through comparative evaluation of robustness, failure modes, and deployment constraints.

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Suwon, South Korea
F-5 Permanent Resident

Languages

- Korean: TOPIK 5
- English: IELTS Academic 7.0
- Bengali: Native

Education

Jeonbuk National University

B.S. Computer Science
and Engineering
2017–Feb 2021

GPA: 3.20 / 4.50

Official university average: 85.10 / 100

Relevant coursework:

- Machine Learning: A
- Artificial Intelligence: A
- Image Processing: A
- Information Retrieval: A

SCHOLARSHIP

Korean Government Scholarship
Program (GKS)
Undergraduate Scholar

Undergraduate Researcher, Data Mining Lab

2018–2019

Supported TREC-related
tweet/event-stream classification
research through data
categorization and linguistic/pattern
analysis of emergency-related
social-media content.

Selected Projects

Adaptive Edge Anomaly Detection for Smart Water Meters

Independent contract / product development | Jan 2026–Present

- Designed and implemented the anomaly-detection method from initial concept through the current product prototype, with edge prototyping on an STM32U5 Nucleo platform before migration to custom production hardware.
- Designed a per-meter adaptive baseline using online mean/variance updates, anomaly scoring, input-validity weighting, and evidence accumulation before an event is raised.
- Designed the baseline update to suppress or down-weight abnormal observations so persistent anomalous behavior is not gradually absorbed into the learned baseline.
- Initially experimented with Isolation Forest and autoencoder approaches, then moved to an adaptive statistical design better suited to household-specific online adaptation and interpretable leak decisions.
- Evaluation on consented household, office, and industrial water-meter data is in progress. Related technical work was incorporated into a patent application filed by the client.

Autonomous Water-Treatment Skimmer | K-water Collaboration

Fluton Inc. | 2021–2023

- Sole software developer for an autonomous surface platform field-tested with K-water engineers.
- Implemented repeatable autonomous cleaning and survey missions using Pixhawk/PX4 and RTK-GPS, with LiDAR-based collision avoidance near pool walls and facility structures.
- Implemented AprilTag-guided docking on NVIDIA Jetson Nano and the post-alignment sequence that engages electromagnetic dock retention and wireless charging.
- Demonstrated complete unattended cleaning, docking, and charging cycles during K-water field testing, including multi-hour autonomous operation.

Robotic Pill-Sorting Vision System

Completed contract project | Jan 2026

- Developed the visual-recognition pipeline for an initial set of approximately 16 pill classes using high-resolution front/back image pairs.
- Trained the initial CNN classifier on project-specific image data, then evaluated DINOv2-based visual representations in later development.
- Designed a confidence-based rejection rule: low-confidence predictions were rejected for manual handling rather than automatically sorted because an incorrect pill decision was considered more costly than a rejection.
- Integrated classification decisions with the robotic sorting controller and developed a retraining workflow for adding new pill classes without redesigning the complete system.

Selected Projects

Lifeguard Autonomous Surface Vehicle

Fluton Inc. | 2021–2023

- Fine-tuned pretrained TensorFlow object-detection models on a project-specific dataset of people in water and deployed them on NVIDIA Jetson Nano.
- Applied MediaPipe pose estimation to detected persons and calculated arm-raise angles from pose landmarks to flag possible distress postures and trigger alerts during survey missions.
- Integrated LiDAR, ultrasonic sensors, and RealSense camera data through ROS2 for obstacle avoidance.
- Connected perception results and vehicle behavior through Pixhawk/ArduPilot, QGroundControl, and custom MAVLink interfaces.
- Used ROS2/Gazebo simulation to test manual-control and autonomous-survey behavior before field trials.

Agricultural Follow-Me Transport Vehicle

Approtec Inc. | 2023–2024

- Combined UWB ranging with camera-based target identification for operator-following functionality.
- Implemented following-distance control and obstacle avoidance.
- Developed BLDC traction-motor and steering-position control software.

Flow-Battery Monitoring and Protection

Approtec Inc. | 2023–2024

- Developed monitoring software for a vanadium redox-flow battery system using CAN and RS485 data for flow, temperature, pressure, and charge/discharge state.
- Implemented configurable threshold-based fault alarms and protective logic.
- Automated inverter control so charge/discharge tests could operate without continuous manual supervision.

Certification

Engineer Information Processing (정보처리기사)

Technical Skills

Programming & Embedded Systems

C, C++, Python, STM32, FreeRTOS/RTOS, Embedded Linux, Yocto, Raspberry Pi, Qt/QML

Interfaces & Control

CAN, LIN, RS485, BLE, serial communication, MAVLink, BLDC motor and steering control

Verification & Development

GoogleTest, CI, QAC, CodeSonar, MISRA/CWE analysis, Git

Professional Experience

ASSOCIATE RESEARCH ENGINEER | GT SYSTEM

Yongin | Jul 2024–Present

- Develop production C++ and Qt/QML software for Hyundai commercial-vehicle instrument clusters from Hyundai/Mobis specifications, including CAN/LIN signal handling and HMI behavior in a Yocto-based environment.
- Perform simulation and physical-device verification, GoogleTest unit testing, CI-based validation, defect reproduction, and on-vehicle testing.
- Review and resolve static-analysis findings using QAC and CodeSonar for MISRA/CWE-related compliance and maintainability.
- Built host-side test automation that drives a physical cluster over ADB, sets hundreds of vehicle configuration parameters, executes tests, evaluates captured display output, and generates reports; reduced a multi-day manual workflow to approximately two hours.

SOFTWARE ENGINEER | APPROTEC INC.

Seongnam | Jun 2023–Jun 2024

- Developed software for the follow-me vehicle and battery-monitoring systems described above.
- Developed Raspberry Pi software for RS485 water-level/flow sensing, automatic pump control, Qt monitoring, and MySQL/Google Cloud SQL data handling.

SOFTWARE ENGINEER | FLUTON INC.

Apr 2021–Jun 2023

- Developed software for the autonomous skimmer and Lifeguard USV systems described above.
- Developed C/RTOS firmware for a shared-bicycle lock system using BLE, GPS, sensor monitoring, and modem communication, including reducing GPS-processing heap usage and resolving watchdog resets during MCU sleep.

ENGINEERING INTERN | H&Y

2021

- Worked with ArduPilot, QGroundControl, and ChibiOS customization for autonomous-platform development.

Robotics & Autonomous Systems

ROS2, Gazebo, Pixhawk/PX4, ArduPilot, QGroundControl, LiDAR, RealSense/camera, ultrasonic, UWB integration, NVIDIA Jetson

Computer Vision & Machine Learning

TensorFlow, OpenCV, object detection, pose estimation, CNN classification, anomaly detection

Working Exposure

PyTorch, TensorRT, ONNX